FYP Progress Meeting

Supervisor: Prof. AN Liang
Reporter: YIP Tsz Yeung Oscar, SEN Bijoy, ZHAO Xuecen
Date: 30, Sep, 2025

THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

Outline

1. Background (Oscar)
2. Personal Task and current progress
   - Bijoy(EEE)
   - Oscar(BME)
   - Jack(ME)
3. Team collaboration (Bijoy)
4. Project outcome (Jack)
5. Project timeline (Jack)
6. Purchase list (Jack)
7. Questions

1. Background

In september, Our group start the experiment in GH701

We collect the data from experiment and plot three sample graphs

Two main parts:

1. Find the best concentration of Vanadyl sulfate solution and H_2SO_4
2. Find the suitable car and design the battery for it
3. Develop the AI model and train it with sample data

2. Personal Task (Oscar)

Analysis the e-fuel performance

Testing we done:

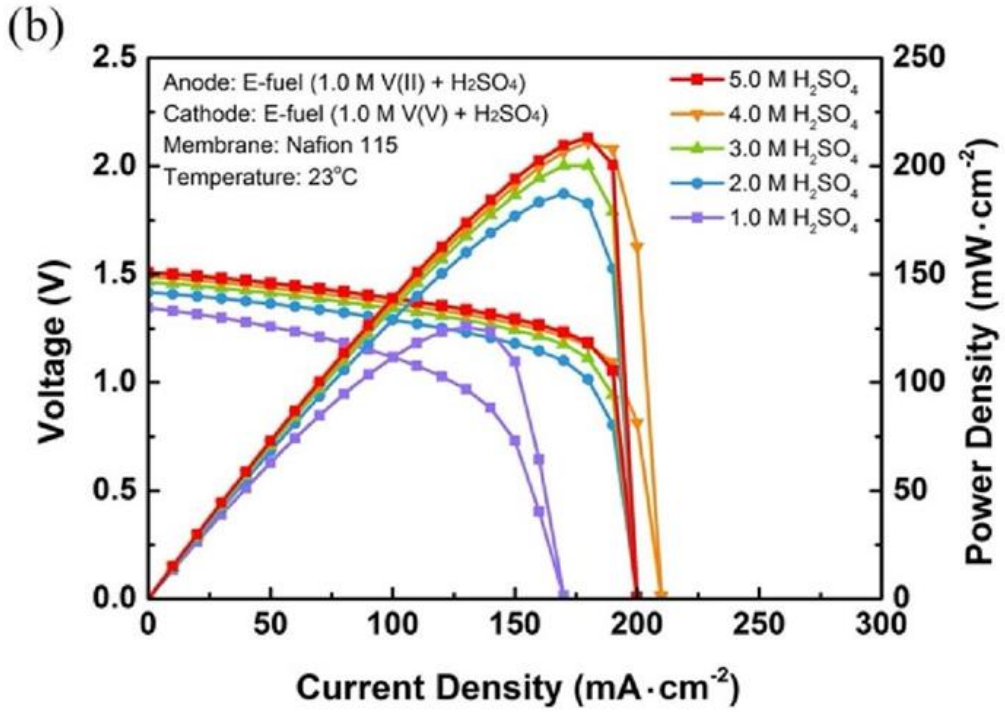
1. 1 M Vanadyl sulfate solution with 2 M H_2SO_4
2. 1 M Vanadyl sulfate solution with 3 M H_2SO_4

Hypothesis result from others article

The article shows the performance of fuel cell

Which we want to follow as guidelines

We will plot the graphs in this format

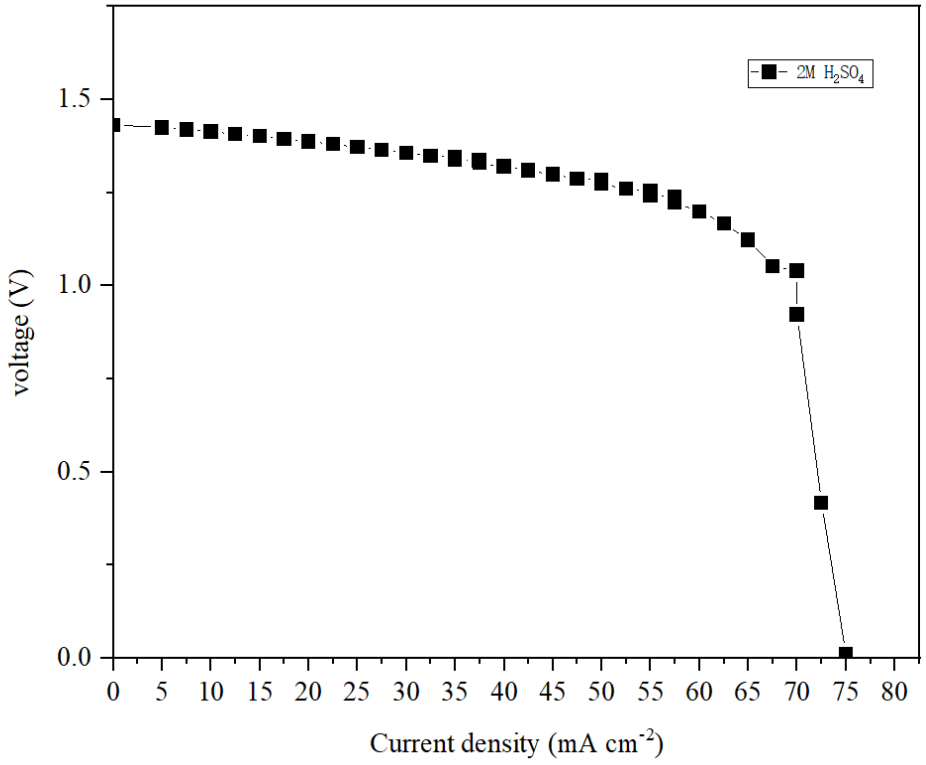


Result of 1 M Vanadyl sulfate solution with 2 M H_2SO_4

The Current density is only **75 mA cm^{-2}**

In the pervious article, the current density should be **200 mA cm^{-2}** which is different with the experiment result

Only the OCV is same when compared with the pervious article

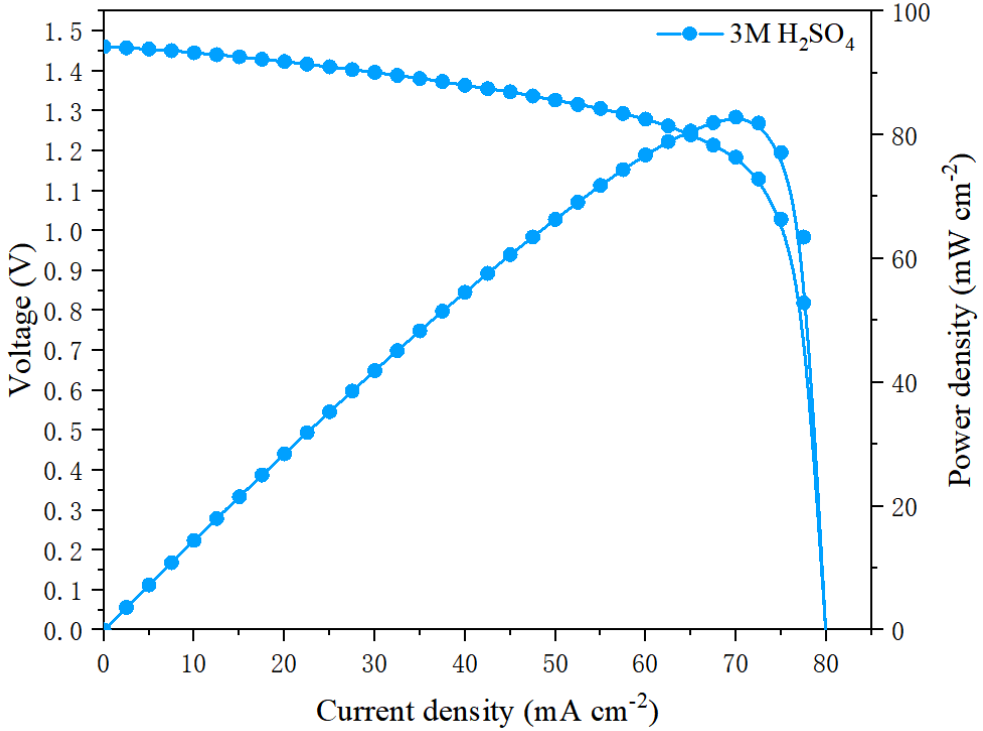


Result of 1 M Vanadyl sulfate solution with 3 M H_2SO_4

Similarly, the Current density is only **80 mA cm^{-2}**

In the pervious article, the current density should be **200 mA cm^{-2}** which is different with the experiment result

Only the OCV is same when compared with the pervious article



YIP Tsz Yeung 22077083D

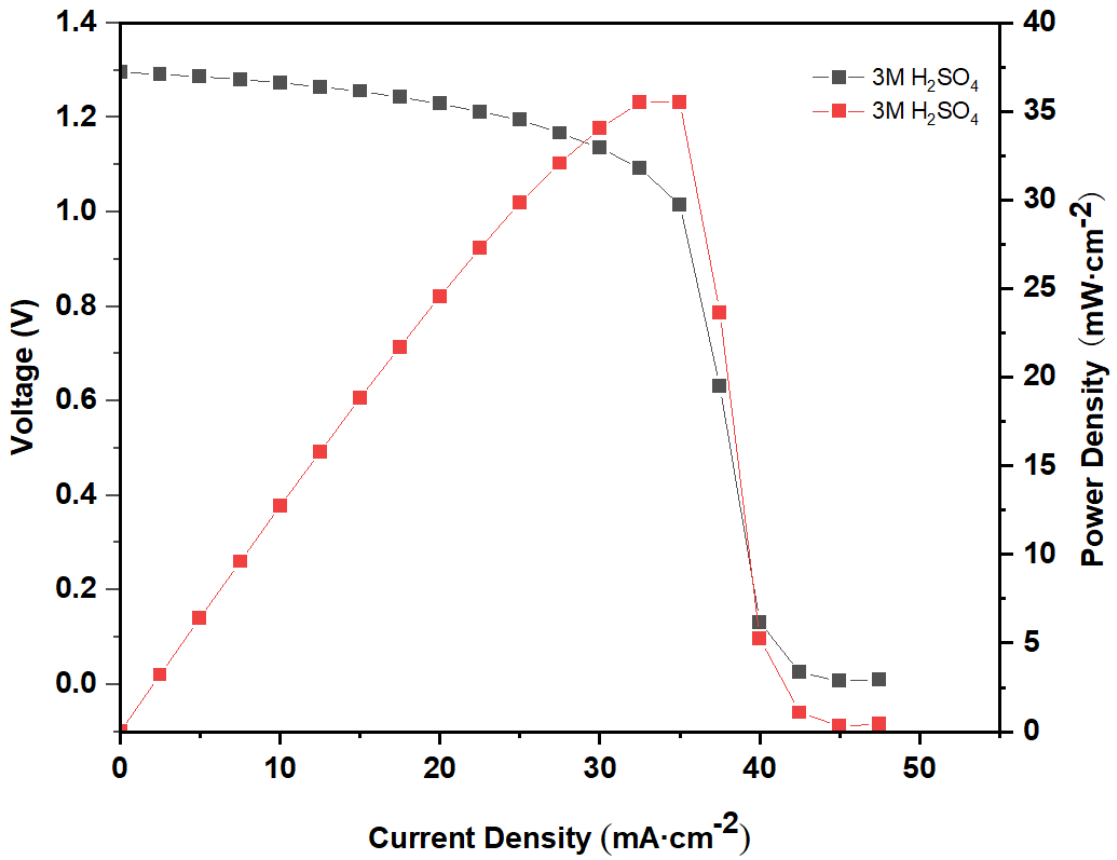
Result of 1 M Vanadyl sulfate solution with 3 M H_2SO_4

This graph is the error result

The error:

1. Add the V^{2+} in V^{5+} chamber
2. Short the circuit during discharge

These kind of mistakes will cause the bad performance of result.



YIP Tsz Yeung 22077083D

How can we improve

Reflection:

> Make sure add the fuel in correct chambers and double check it before fill in

> During discharge, make sure the circuit is not shorter, which check the OCV from the software and make sure OCV is not zero

Take care the chemical and finish the discharge process as fast as possible.

22077083D

Hypothesis of bad performance

In the first charging process :

V^{5+} is **yellow** in colour

In the second charging process:

The V^{5+} bottle show in **dark green** colour

The setting of second charging in flow cell may have problem

2. Personal Task (Bijoy)

> **Assemble the overall car:**

- Use the components (to be bought), to assemble the internal components of the car, which will include:
  - Electrical parts: Jetson Orin, Binocular Camera, Wifi Module, and BMS
  - Mechanical parts: Brushles Motors, Servor Motor for direction, Car Suspensions, and Motor Driver Modules

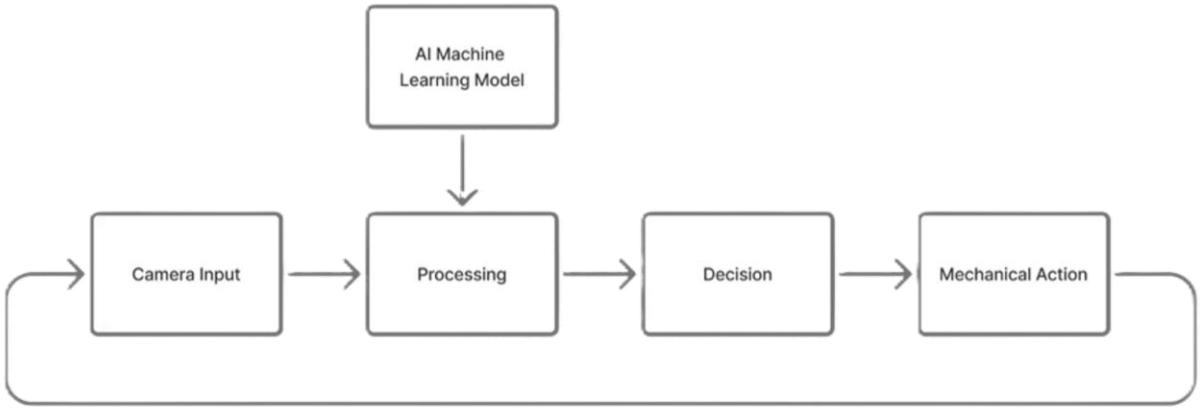
> **Start collecting sample data:**

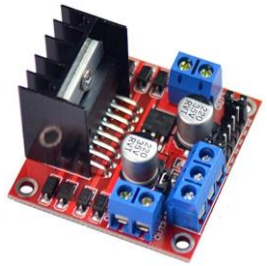
- Collect sample data of traffic signals and road-signs using the camera mounted on the car to use as training data for the AI

> **Begin building the AI model:**

- Start coding the AI architecture using Python and an open-source model (from hugging face) which will be used to detect road signals for the car

Electro-Mechanical Architecture





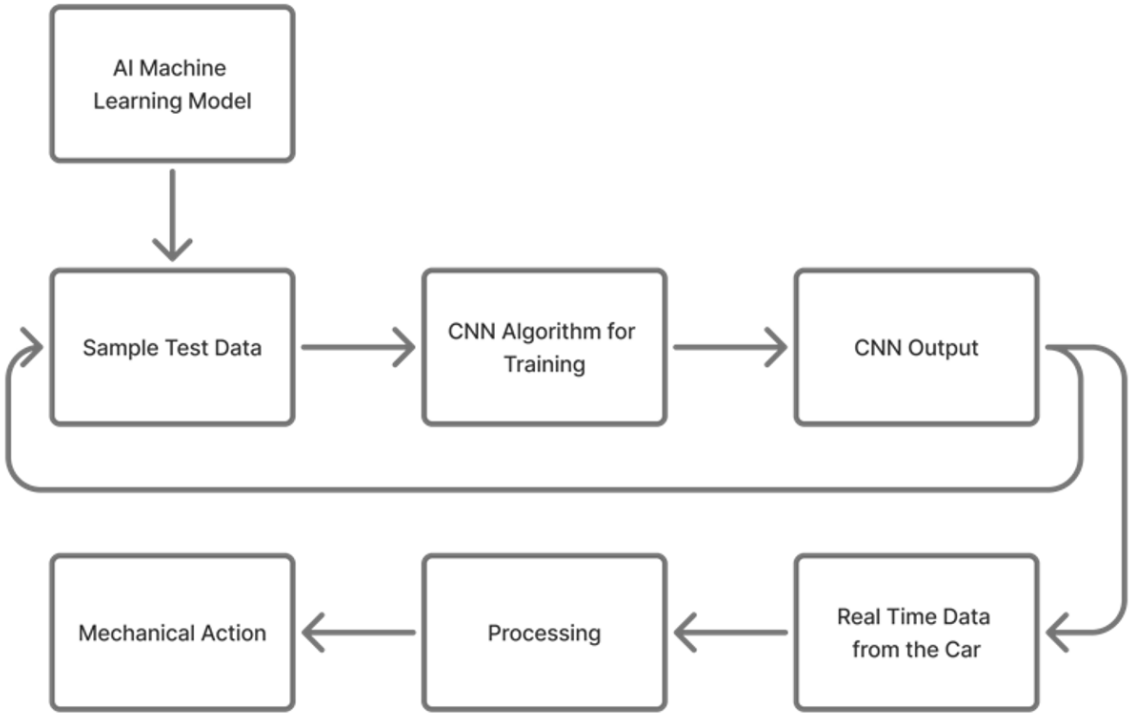
> High Level Electrical Design

- The car's main brain will be the Jetson Orin, which is an NVIDEA microprocessor for AI applications
- The Jetson Orin will host the machine learning model locally to be used as navigation

> High Level Mechanical Design

- The car will use 4 brushless motors with 1-3k RPM speed (with dynamic speed control) - Vector value (-1 to 1)
- The car will also use a servo motor to control direction - Vector value (-1 to 1)

AI Architecture



> AI Model and Algorithm

- The AI model to be used will be an open-source model and will be trained on the algorithm called Convoluted Neural Network (CNN), which is perfect for image-based machine learning

> AI Model Training

- We will be using the NVIDEA RTX 5070ti and utilize its CUDA cores to train the model with sample data taken from the car's camera
- We will be using a Mini-PC to host the model in the training phase, and then transfer to the Jetson Orin while using it for the car
- We will use the trained model and then use test data to confirm if it can detect traffic signals

AI Architecture

```python
import tensorflow as tf
from tensorflow.keras import layers, models, datasets

# 1. Load and prepare the dataset
(train_images, train_labels), (test_images, test_labels) = datasets.mnist.load_data()
train_images, test_images = train_images.reshape((-1, 28, 28, 1)), test_images.reshape((-1, 28, 28,
1))
train_images, test_images = train_images / 255.0, test_images / 255.0

# 2. Build the CNN model
model = models.Sequential([
    layers.Conv2D(32, (3, 3), activation='relu', input_shape=(28, 28, 1)),
    layers.MaxPooling2D((2, 2)),
    layers.Flatten(),
    layers.Dense(10, activation='softmax')
])

# 3. Compile and train the model
model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

model.fit(train_images, train_labels, epochs=1)

# 4. Evaluate the model
test_loss, test_acc = model.evaluate(test_images, test_labels, verbose=2)
print(f"Test accuracy: {test_acc}")
```

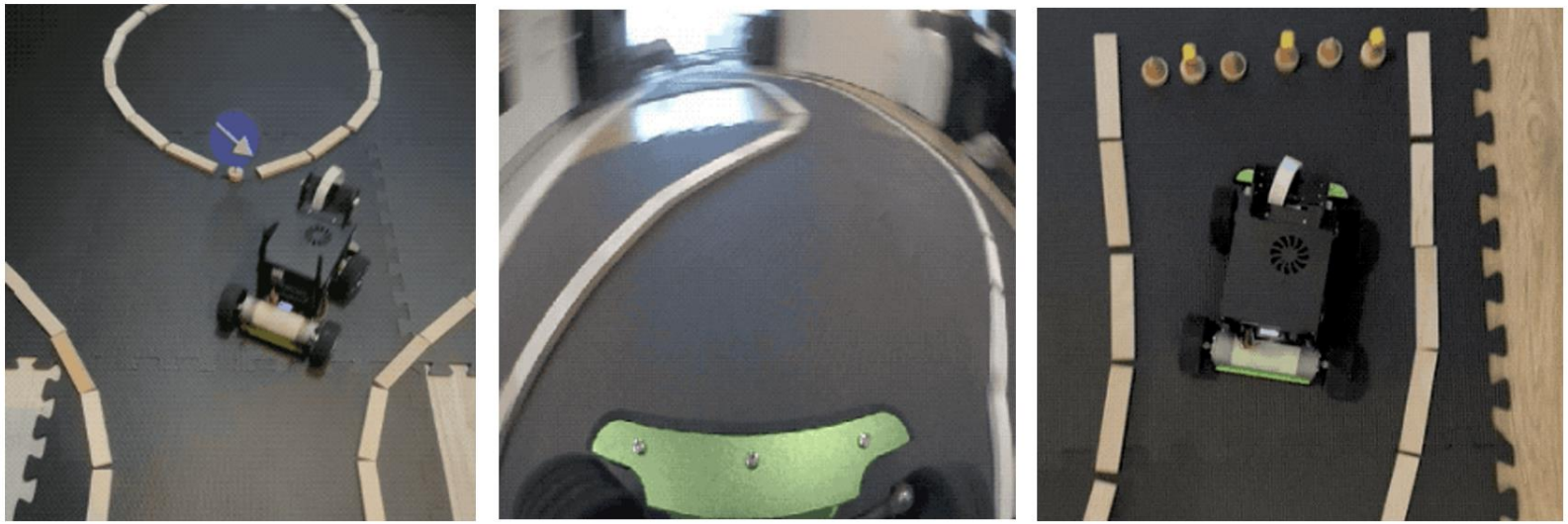
> AI Code

- The figure shows a really small example code snippet of how a CNN algorithm will be used
- The model uses multiple layers to determine different factors in an image and learn from it

> AI Model Training

- We will use the "train_image" and "train_label" functions to first label the image data and then train on the labelled data using supervised learning

AI Architecture – Sample Data



> **Sample Data**

- The figure shows some sample data which can be used to train the model
- The traffic signals and roadside signs like white lines, turn signals and blockades are all used for training

> **AI Model Usage**

- The trained model will then be used for the car to be able to drive autonomously

2. Personal Task (Jack)

> **Design the passive fuel cell stack** using CAD tools for mechanical optimization. This includes designing the anode, cathode, membrane, current collectors, and flow fields.
  - Stack is expected to be capable of Open-circuit voltage of 1.4 V, a maximum current of 2.5 A, and a peak power of 2.0 W, running time for a minimum of 10 minutes ≈ 90 mAh or 0.333 Wh.

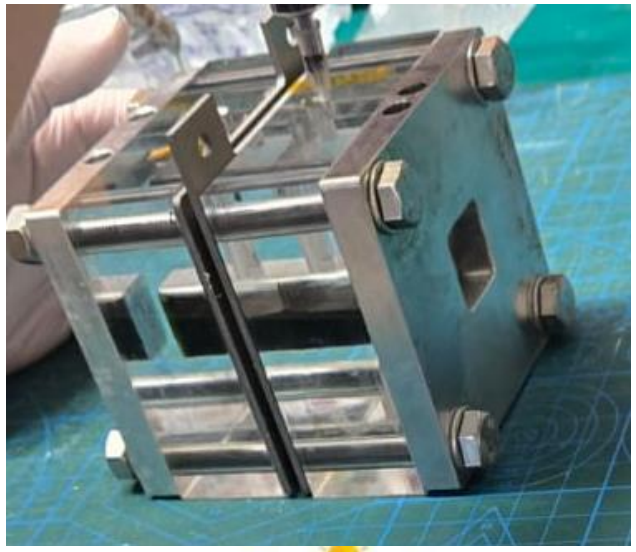
> Collaborate closely with our BME teammate to **calculate the suitable size and reaction area** of the fuel cell.

> **Fabricate fuel cell components** using processes such as CAD design, CNC machining, and assembly, while ensuring the use of suitable materials.

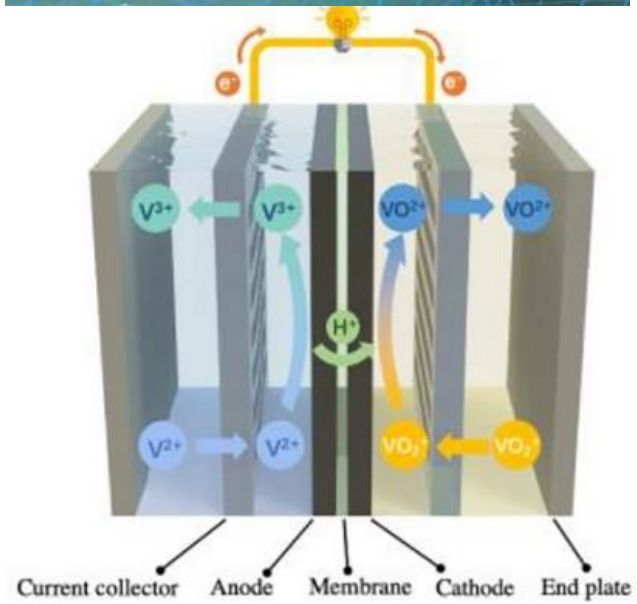
> **Integrate the fabricated components** into a functional stack prototype and support team efforts in its evaluation and assembly into a toy car.

> **Train the AI model** in collaboration with our EEE teammate.

Provided lab fuel cell design

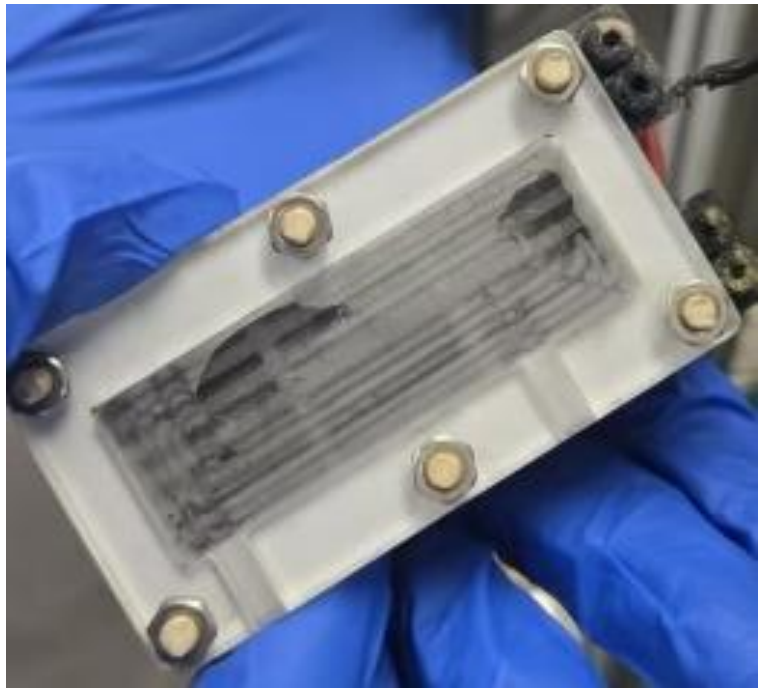


- It is way **too big and heavy** for the car and only suitable for reference and experiment.
- Assembled and disassembled to learn the **configuration** of the design and **working principle** of passive fuel cell.
- From refueling and cleaning this design, we understand a lot more about the passive fuel cell's **working process** and helped me **brainstorm ideas**.
- We also going to **test e-fuel performance at different concertation** with this design before we make our own design.



Previous FYP fuel cell design analyses

- **Compact design with sufficient power for its purpose**: The design is simple yet efficient, but for our project, we will require a much higher energy. I need to think of a design that's capable for our demanding system.
- **Difficult Refueling Process**: The extremely small refueling openings make the process challenging, a problem that will limit the design's scalability with a larger stack.
- **Insufficient Fuel Capacity**: The tank's small size, combined with the poor power density of e-fuels, will result in a very short runtime for the battery for our more power intensive system.
- **Prone to Refueling Errors**: The concentration of all fuel fillers on one side of the battery makes it easy to accidentally mess up the refueling process.
- **Challenges in Removing Spent Fuel**: The design makes it difficult to drain or remove used fuel from the system.
- **Risk of Fuel Spillage and Corrosion**: The lack of a cover on the fuel filler poses a significant risk of fuel spilling out, which could corrode other components.

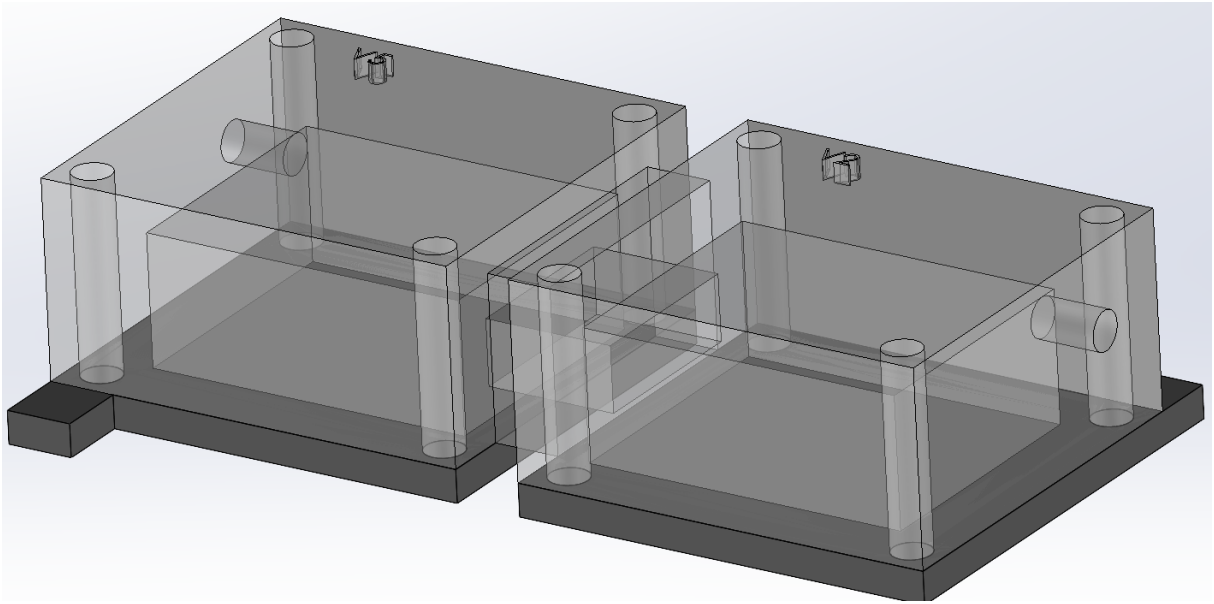


Potential solution to the problem of the existing design

- **Easier Refueling Process**: Extra tube that connect the cell stacks for easier fuel delivery.
- **Clear fuel logo**: Mark polarity clearly on the cell.
- **Expand Fuel Capacity**: Make sure the tank have sufficient room for fuel storage.
- **Easier Fuel removal**: The fuel can be easily removed from the tube mentioned in part one.
- **Plug to contain the fuel**: A simple plug to avid spilling when driving.

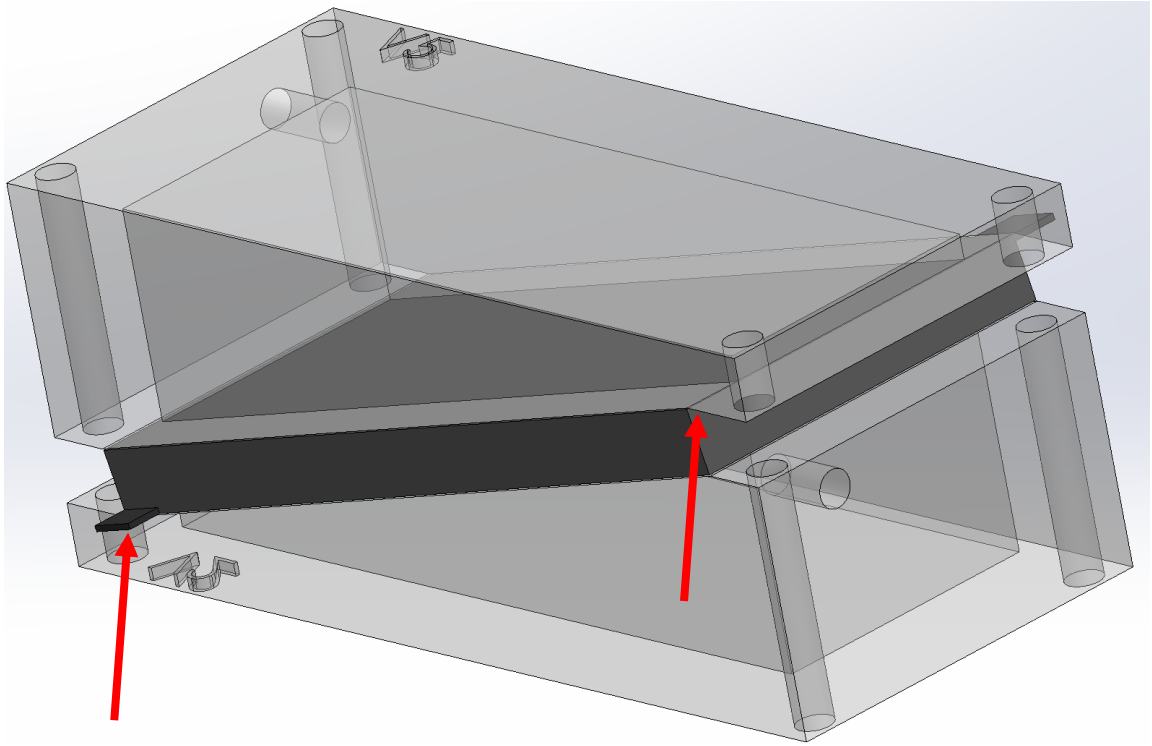
Conceptual design one

- A design that utilize the fact that the current collector does not have to be next to the membrane.
- Easily **stackable** (vertically and horizontally) and have **great capacity**.
- Current collector is placed at the bottom, which means the **reaction area** is also greater.
- Current collector at the bottom witch can utilize the movement of the car while driving for **better convection**.
- However, the design may face **critical efficient issue** as the membrane is separated from the current collector.



Conceptual design two

- A slanted design.
- Easily **stackable** (horizontally) and have **great capacity**.
- Current collector is slanted, which means **greater reaction area** compared to original.
- Current collector is sandwiched between the two tanks so there won't be efficiency problems like design one.
- However, the design may still face critical problem as the acrylic tank may break easily at the **weak spot**(mark with red arrow). The special shape also make it **hard to manufacture** and **potential leakage** at the slant.

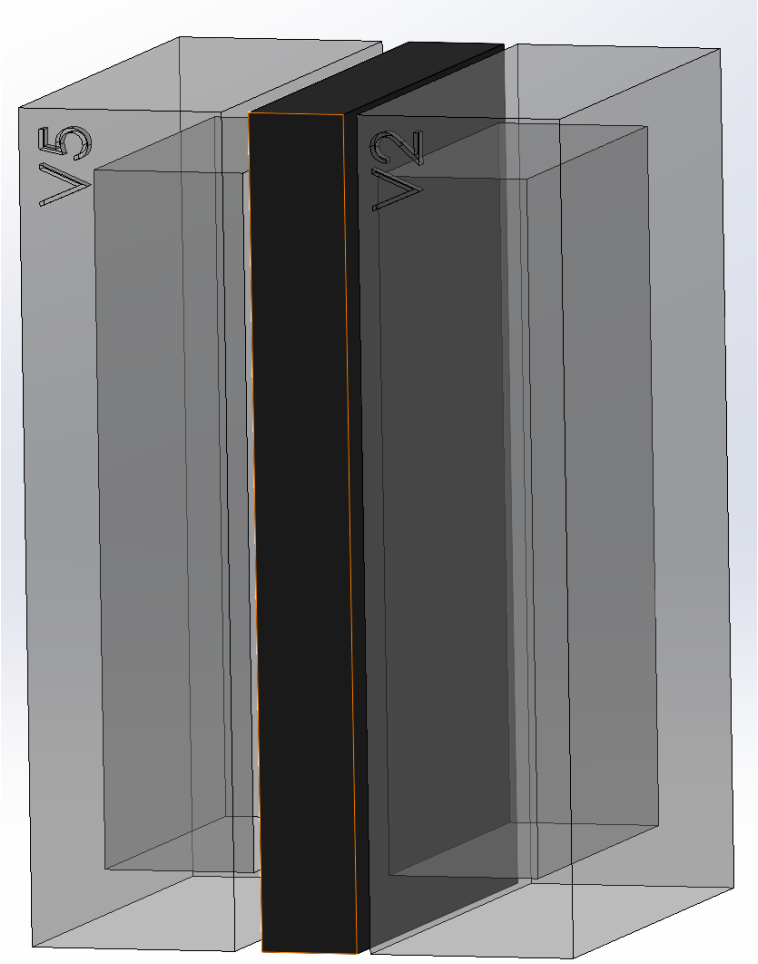


Other ideas I have but seems too difficult or unrealistic

- **Pressurized fuel tank** chamber with hand pump that will drop the fuel slowly through the fuel cell by gravity with a waste tank for spent fuel.
- **Curved current collector** for higher reaction area.
- Stack with **a few current collector in one singular tank** for each node.
- Semi passive fuel cell with **pump powered by solar panels**.
- **“回” shaped design** with inside V5, outside V2 and the inner wall being 4 current collector.
- **Curved fuel tank interior** for better convection.

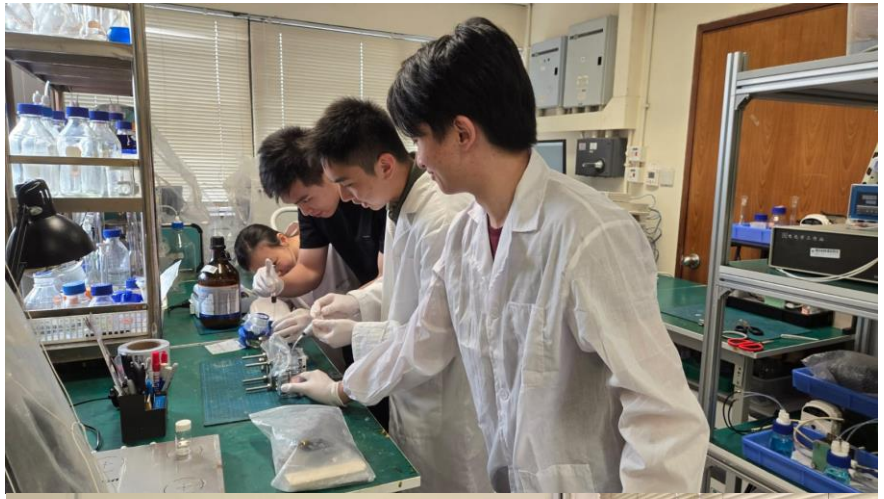
Why do I try to make these design?

- To address the refueling problem, I suggest the hole for refuel are all located at one side(left for V5, right for V2).
- If the current collector stand up right, we need to sacrifice either reaction area, stack ability, size or fuel capacity.
- If the current collector is sandwiched and laid flat, the bottom liquid will not be reaching the current collector.
- They are **concept designs**; I still have many things that I need to learn and making simple conceptual design in SolidWorks can be useful at **brainstorming** new ideas and **identifying potential problems**.
- I want to show and make discussion on the concepts before making a more detailed, complete design.
- Seek professional advises.



3. Team collaboration

- **Working closely with the international team**: Since we share the same set of equipment and are doing similar experiment, we share our lab experience and compare our data for better accuracy.
- **Have regular online and offline meeting**: We discuss our progress and share our opinion on the project with meetings.
- **Always use WhatsApp for effective communication**: When we encounter problems and issues, we always contact each other to seek solutions and arrange meetings if necessary.



4. Project outcome

E-Fuel Cell System

> **Performance:** 1.4V open-circuit voltage, 2.5A max current, 2.0W peak power.

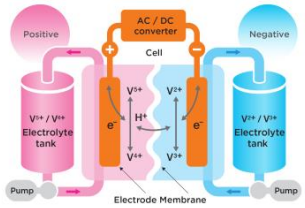
> **Duration:** Must run for at least 10 minutes.

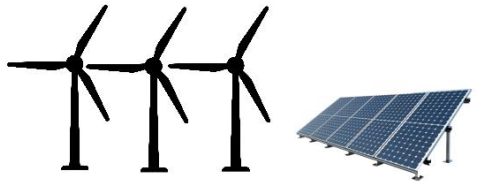
Autonomous Car & AI Showcase

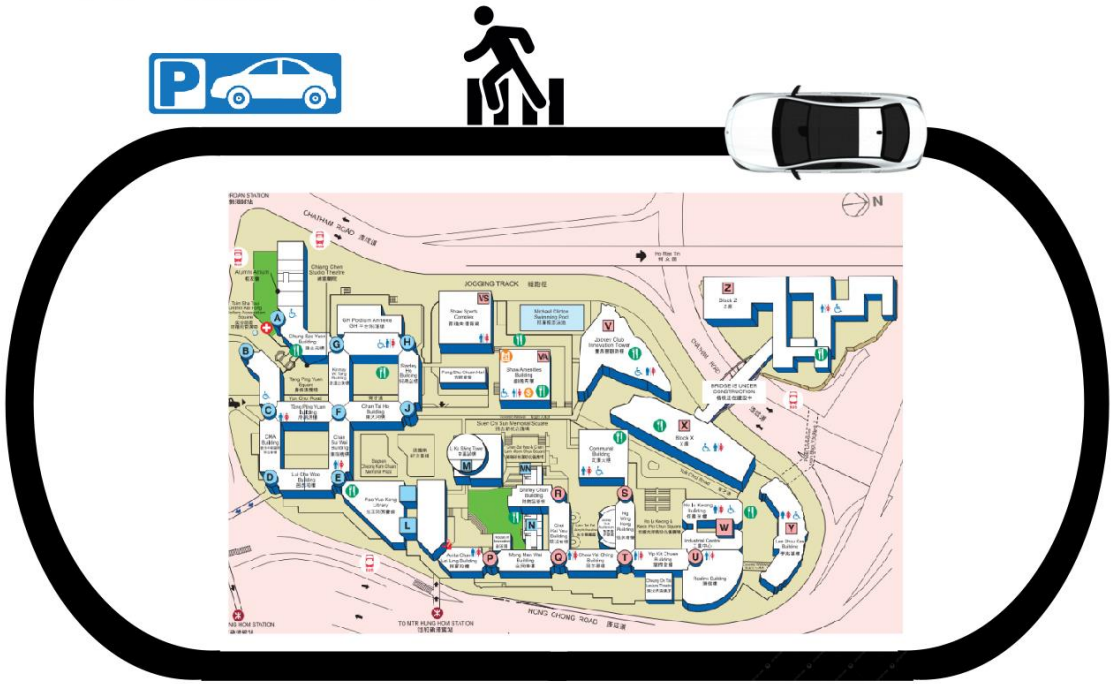
> **Car Features:** Integrated with sensors and a navigation system.

> **AI & Software:**
- An AI model, trained with YOLO (You Only Look Once), handles object detection.
- The AI model runs locally on a PC using Docker.

> **Demonstration Track:**
- A simple automatic drive track to showcase the AI.
- Features a blueprint map of the PolyU campus.
- Includes interactive elements: traffic lights, parking spaces, and crosswalks with passing pedestrians.







Design and Development of an Electric Toy Car Powered by a Fuel Cell Stack

4. Project outcome… if time and budget permits

> Construct a **3D-printed model** of the PolyU campus to replace the blueprint.

> Enhance the demonstration with voice control capabilities by integrating an **open-source Large Language Model** (LLM) hosted on Ollama.

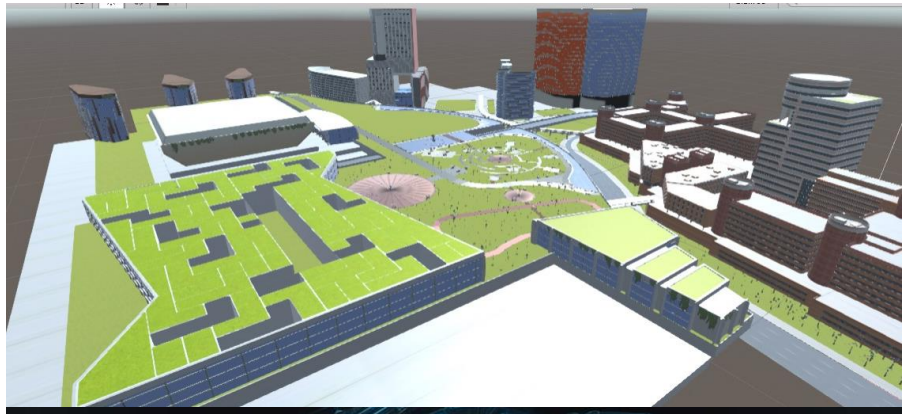
> Utilize **ComfyUI** to manage the interaction pipeline.

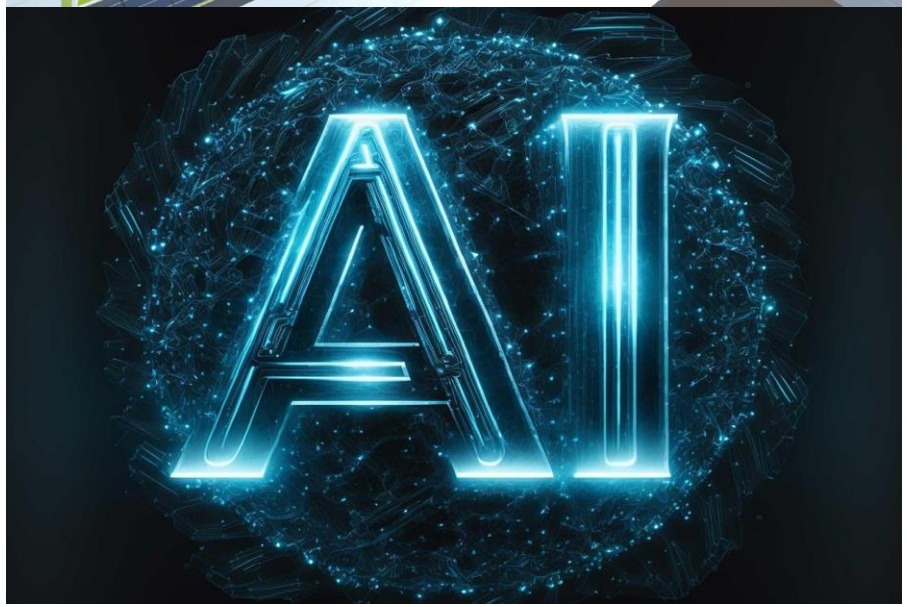
> The system will be able to **respond to verbal questions and commands**, such as:

- "Tell me more about the future of Vanadium batteries."
- "Who designed this project?"
- "Please go back to the parking station."

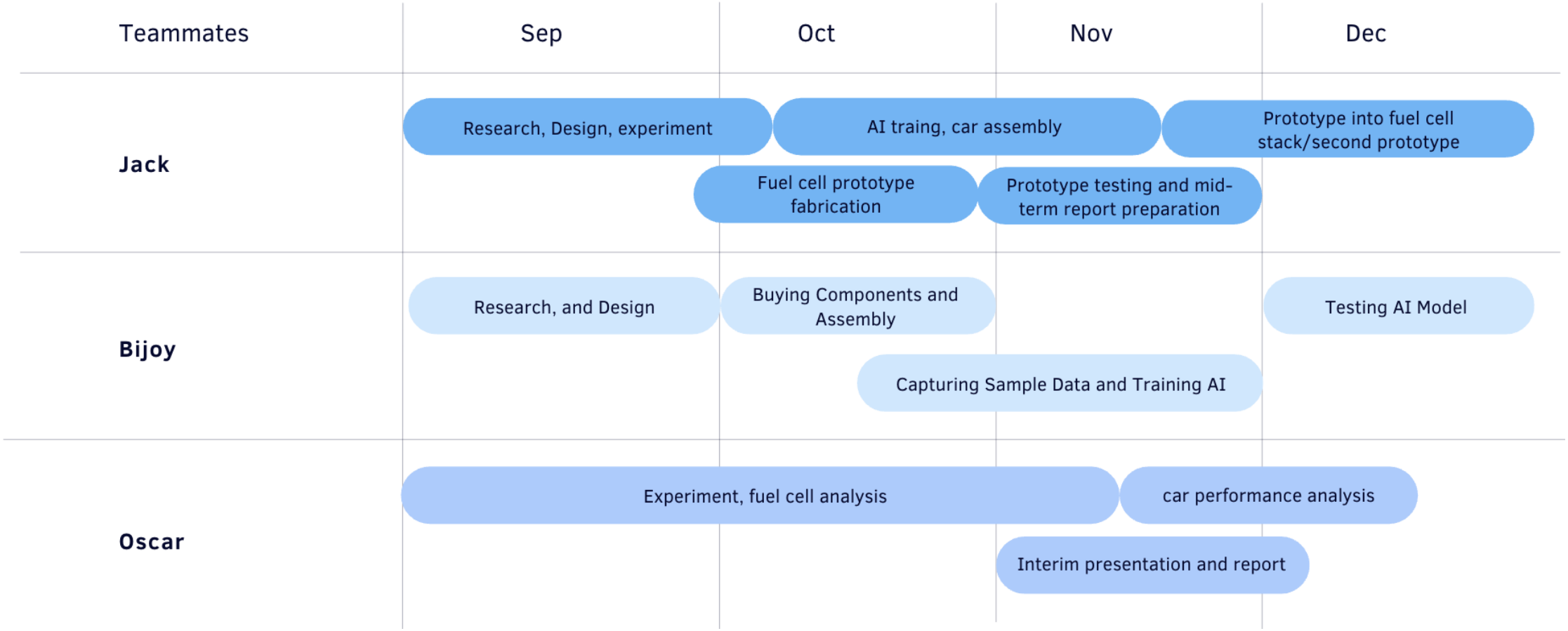
> Implement a hardware setup with a microphone at the demonstration stand to capture voice input.

> Process the audio input on a PC, which will generate commands for the LLM and then relay actionable instructions to the toy car.





5. Project Timeline Chart



SEN Bijoy 22097037d

6. Purchase list

Product Name	Amount	Price HKD (Appx)	Total
Battery Fabrication	x	x	10000
Jeson Orin/Nano 8gb	1	4000	4000
AI Mini PC	x	5000	5000
RTX 5070ti 16 GB	1	7200	7200
Binocular Camera	1	1500	1500
High Speed WIFI Module	1	500	200
Car Body	1	2000	2000
Mechanical Components	x	x	3000
Electrical Components	x	x	3000
Multi-Purpose Tools	x	x	1000
Demo Components	x	x	2000
Software Licences	x	x	1000
Total			39900

Top AI performance with low-power efficiency in a small form factor for embedded and edge applications

Featuring fifth-generation Tensor Cores for enhanced AI performance, alongside 16 GB of high-speed GDDR7 memory

Ability to directly capture stereoscopic information by mimicking human vision, which allows for accurate distance and spatial reasoning

7. Questions

- How would the demonstration take form? (duration, location, are we going to be there the whole time?)
- For the Interim Report and presentation, what will be the requirements or expectation?

Thanks for your listening !